Do Options Overprice Macro News?

SPY options around CPI, payrolls and FOMC decisions, 2022–2026

Unai Fueyo Montalvo · BSc Economics, Universidad Autónoma de Madrid November 2026

The question. Options usually charge more for protection than the moves turn out to be, the volatility risk premium. Does it survive on the days the market fears most, CPI, payrolls and Fed decision days? Can it be earned after costs, and when within the day does the event risk leave the price?

1,121 days
VIX1D's priced move vs the S&P 500's actual move, May 2022 – Nov 2026

267 straddles
real SPY options expiring the next day, 1-minute prices, Oct 2025 – Nov 2026

3×10^{-5} vol pts
largest gap between my Black–Scholes pricer and the vendor, 46,816 option-minutes

26 tests
7 scripts, one command reproduces every table and chart

SIX FINDINGS

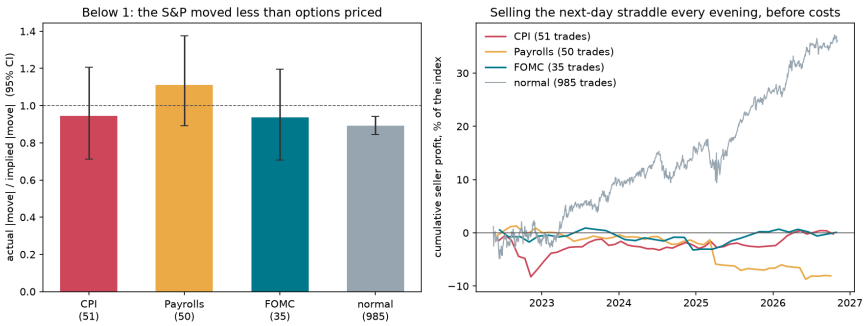
1. **Normal days are overpriced.**
   The S&P moves 0.89 times what options priced [95% CI 0.84, 0.94] on 985 days; a straddle seller wins 62% of days. Robust to every check.
2. **On release days the premium shrinks.**
   Options charge 27% more than on normal days, but the index moves 43% more (medians: 26% and 44%). Consistent on every check, not significant overall (p = 0.19).
3. **Payrolls is the event options miss.**
   Overnight, options price almost no extra risk for the jobs report, yet the move is 1.5 times what was priced (2022–26: 1.11, also without the 2025 tariff crash).
4. **CPI flipped.**
   Too cheap in the 2022 inflation shock (1.48), too expensive since 2023 (0.78 [0.60, 0.96]).
5. **Skip the releases.**
   After retail costs, selling every evening has a Sharpe ratio of 0.23 and a 16% drawdown; skipping release days, 0.52 and 8.6%.
6. **The Fed's risk leaves during the press conference.**
   Just after the 14:00 statement 89% of the priced variance remains; at 15:00, 49% (normal days 73%).

ACTUAL OVER PRICED MOVE

Day	2022–26	2023–26	Overnight*
CPI	0.94	0.78	0.73
Payrolls	1.11	1.17	**1.50**
FOMC	0.94	0.91	0.62
Normal	**0.89**	0.88	1.04

* eve close to 09:35, real SPY options, Oct 2025 – Nov 2026.
Below 1: options charged too much.

Validate before you estimate. Put–call parity was off by 11 cents: the interest on the strike over the T+1 settlement gap, three days on Thursday eves, plus SPY's dividend; after both, 2 cents. Half days close at 13:00, and the pricer's clock knows it. VIX1D was checked against real straddles (correlation 0.989). The 2025 tariff crash, labelled a payrolls day, is kept: every robustness check applies one rule to all days.



Findings 1–2. Left: actual over implied absolute move by kind of day, May 2022 – Nov 2026 (95% CI). Right: cumulative profit of selling the next-day straddle every evening, before costs: steady on normal days, flat or negative on release days.

WHY IT MATTERS

Selling options earns a premium, but not on the days investors fear most. A seller who avoids macro releases keeps the premium with half the drawdown, and payrolls, not CPI, is now the risk the options market underprices.

READ MORE

Full paper, code and tests on GitHub. Data: Cboe, FRED, London Strategic Edge. Pricer written from scratch in Python, with AI coding assistance; design and interpretation are the author's. Companion study: Prediction markets and the Fed

For research and educational purposes only; not investment advice. Straddle = at-the-money call + put expiring the next day. CI = 95% bootstrap confidence interval.